

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 10: Arithmetic Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of n terms

1. An A.P. has first term 5 and last term 50, and its terms add up to 440. Find (a) the number of terms (b) the common difference. **ICSE 2025, 3 marks**

SOLUTION

$$(a) S = \frac{n}{2}(a + l): 440 = \frac{n}{2}(5 + 50), \text{ so } n = \frac{880}{55} = 16.$$

$$(b) l = a + (n - 1)d: 50 = 5 + 15d, \text{ so } d = 3.$$

Answer: (a) 16 terms (b) common difference 3

2. The n th term of an A.P. is $T_n = 6(7 - n)$. Find (a) its first term and common difference (b) the sum of its first 25 terms. **ICSE 2024, 3 marks**

SOLUTION

$$(a) T_1 = 6 \times 6 = 36 \text{ and } T_2 = 6 \times 5 = 30, \text{ so } a = 36 \text{ and } d = -6.$$

$$(b) S_{25} = \frac{25}{2} [2(36) + 24(-6)] = \frac{25}{2} \times (-72)$$

Answer: (a) $a = 36, d = -6$ (b) $S_{25} = -900$

2 The general term

3. Find the 7th term of the A.P. $\frac{1}{a}, \frac{1}{a} + 1, \frac{1}{a} + 2, \dots$ (a) $\frac{1}{a} + 6$ (b) $\frac{1}{a} + 7$ (c) $\frac{1}{a} + 8$ (d) $\frac{1}{a} + 7^7$

ICSE 2024, 1 mark

SOLUTION

$$\text{First term} = \frac{1}{a} \text{ and } d = 1. T_7 = \frac{1}{a} + 6 \times 1$$

Answer: (a)

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.